Name	Subject Class	Class	Candidate Number
	2BI		

ANGLO-CHINESE JUNIOR COLLEGE
Preliminary Examination 2014

BIOLOGY **9648/02**
25 AUGUST 2014
HIGHER 2 **2 hours**

Paper 2 Core Paper

Additional Material: Writing Paper

READ THESE INSTRUCTIONS FIRST
Write your name, index number and class on this answer booklet.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.

Section A
Answer **all** questions.

Section B
Answer any **one** question.

At the end of the examination, circle the number of the Section B question you have answered in the grid opposite.
Fasten all your work securely together.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
7	
8 or 9	
Total	**100**

This question paper consists of **18** printed pages.

[Turn over

For examiner's use

1 (a) Fig. 1.1 below is an electron micrograph of part of a human liver cell.

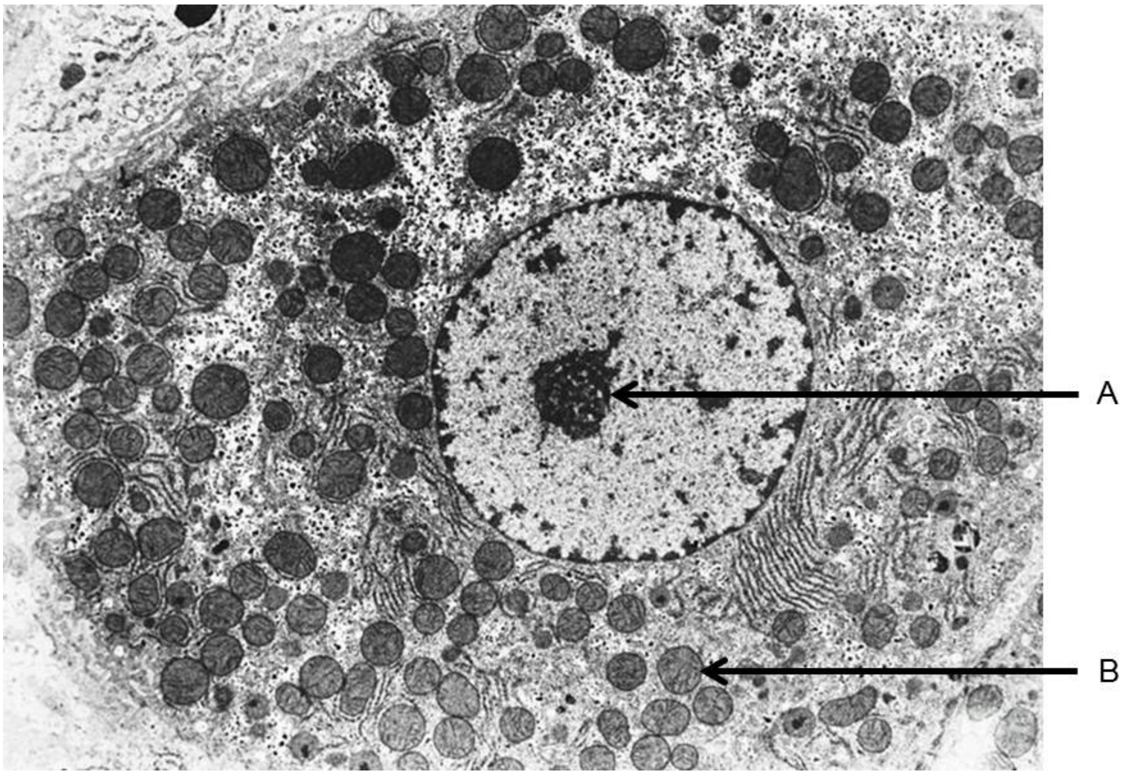


Fig. 1.1

(i) With reference to Fig. 1.1, describe the structure of **A** and state **one** difference in the roles of structures **A** and **B** in the cell.

1. It is a non-membrane bound / dense / region containing DNA coding for rRNA; A! Any 2 out of 3

2. A synthesizes rRNA/ assembles ribosomal subunit vs B produces ATP; R! forms ribosome

@ 1m

[2]

(ii) Suggest why there is a large number of **B** in the liver cell.

1. Synthesis of ATP for glycogenesis/ glycogenolysis/ gluconeogenesis/ production of bile/ iron storage/ protein (hormone) synthesis/ deamination of amino acids/ conversion of ammonia to urea / detoxification;

@ 1m

[1]

For examiner's use

(iii) A large number of storage polysaccharide is found throughout the liver cell. Explain how the structure of this storage polysaccharide makes it a good energy storage molecule.

1. Glycogen is composed of several hundreds or thousands of glucose molecules (linked by glycosidic bonds);
2. Upon hydrolysis, the large number of glucose molecules, which are the main respiratory substrates/ are released to produce energy;
3. Most of the hydrophilic hydroxyl groups of the glucose residues project into the interior of the helices;
4. Hence glycogen is insoluble in water (and can be stored in large quantities) without affecting the osmotic potential of cells / do not easily diffuse out of cells;
5. Glycogen is (highly) branched;
6. Thus allowing it to be compacted, so that more glycogen can be stored in a given space / greater amount of glycogen can be stored per unit volume/ allows many enzymes to act on it at any one time, (hence can be easily hydrolysed to release glucose);

@1m, max 4 (correct structure linked to function)

[4]

(b) It is common for patients suffering from liver disease to have symptoms such as jaundice, abdominal pain, swelling and itchy skin. To reduce the pain caused by the disease, doctors usually prescribe painkillers to the patients. A common type of painkiller is Acetaminophen and it works by blocking calcium ion channels at synapses between neurones.

(i) Explain why there is a need for calcium ion channels to transport calcium ions across the membrane.

1. Calcium ions are charged, hence hydrophilic; R! polar
2. Unable to pass through hydrophobic core / hydrocarbon tails of the membrane / phospholipid bilayer;

@ 1m

[2]

(ii) Explain how the blocking of calcium ion channels can result in the reduction of pain in patients with liver disease.

1. No influx of calcium ions into presynaptic terminal, no movement / fusion of synaptic vesicles to presynaptic membrane;
2. No exocytosis of neurotransmitter, no binding (of neurotransmitter) to receptor on postsynaptic membrane;
3. No opening of chemically gated Na^+ channel, no influx of sodium ions;
4. *No depolarisation of postsynaptic membrane, hence no transmission of nerve impulse/generation of AP;

@ 1m, *compulsory [3]

[Total: 12]

2 **(a)** In 1882, the German botanist T.W. Engelmann performed an ingenious experiment to investigate the effects of different wavelengths of light on the rate of photosynthesis for a suspension of alga *Spirogyra* (a filamentous microorganism containing long, spiral chloroplasts).

In his experiment, a prism was placed between the light source and the alga filament to produce and scatter all colours of the light across the alga filament evenly. Then, aerobic bacteria were added to the alga filament suspension. All the other variables were kept constant.

After exposure to light for a certain period of time, the aerobic bacteria were found to move towards and accumulate at specific lengths along the alga filament as shown in Fig. 2.1. Engelmann concluded that the product released during photosynthesis must have caused the directional movement of aerobic bacteria.

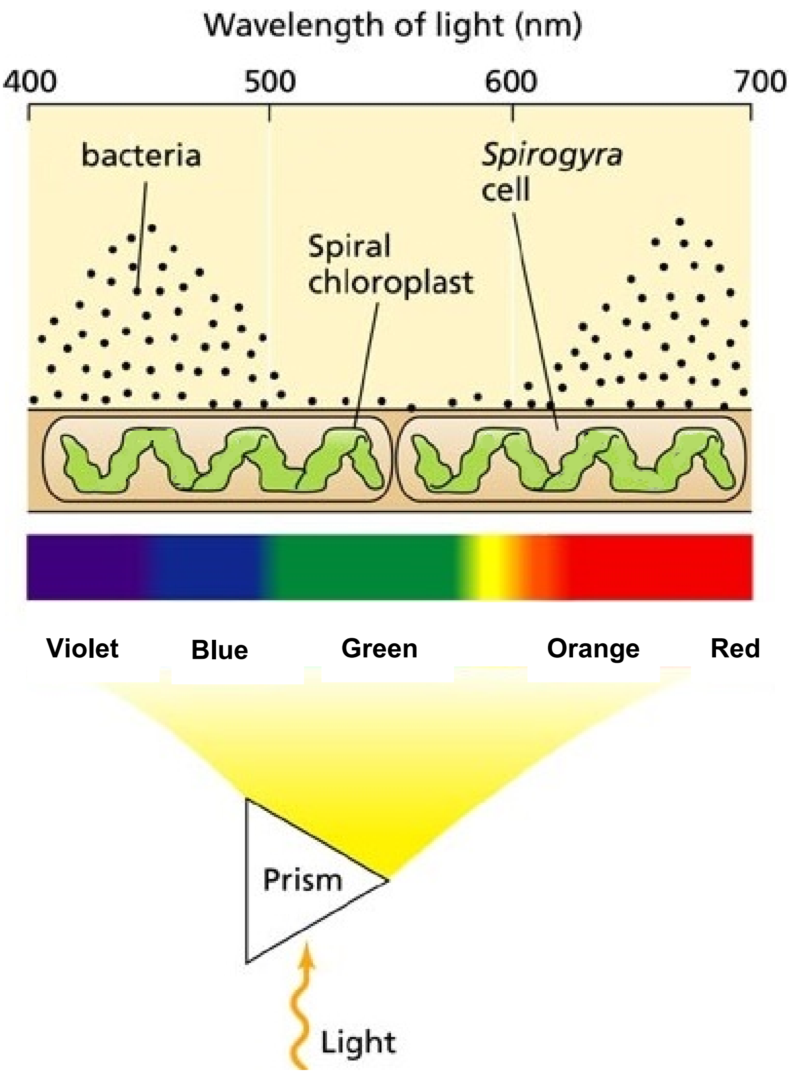


Fig. 2.1

With reference to the information above and Fig. 2.1,

(i) suggest how the rate of photosynthesis was determined in this experiment.

Ref. to quantify aerobic bacteria found at a specific region of alga filament;

@ 1m [1]

(ii) explain why the rate of photosynthesis can be determined as suggested in part (i).

Rate of photosynthesis correlates with the rate of oxygen produced, which is used by aerobic bacteria for respiration;

@ 1m [1]

(iii) describe the relationship between the rate of photosynthesis and the wavelengths of light.

1. **The rate of photosynthesis is <u>higher</u> in the alga filament exposed to wavelength of 400 to 500 nm and 600 to 700 nm as shown by the higher numbers of bacteria found;**
2. **The rate of photosynthesis is <u>lower</u> in the alga filament exposed to wavelength of 500 to 600 nm as shown by the lower numbers of bacteria found;**

The rate of photosynthesis is <u>highest</u> in the alga filament exposed to the wavelength of (450-455) nm and (670-675) nm.

@ 1m [2]

(iv) account for the rate of photosynthesis observed for the alga filament exposed to the wavelength of light at 550 nm.

1. **Ref. to green light is reflected/least/less absorbed by the chlorophyll / photosynthetic pigments in the alga chloroplasts;**
2. **Ref. less photoactivation (of chlorophylls) to transfer the electrons down the electron carriers (in the ETC during photophosphorylation);**
3. **Less photolysis of water and therefore less oxygen evolved;**

@ 1m [3]

(b) The directional movement of cells towards a specific chemical stimulus is known as chemotaxis. It plays critical roles in regulating physiological processes such as the migration of white blood cells to sites of infection and metastasis of cancer cells. The migration of cells involves the rearrangement of a type of cytoskeleton known as actin filament.

Fig. 2.2 illustrates a model of the chemical signalling involving G-protein coupled receptor (GPCR) and different cellular proteins (Dock, Elmo, Rac), which occurs during the migration of most eukaryotic cells.

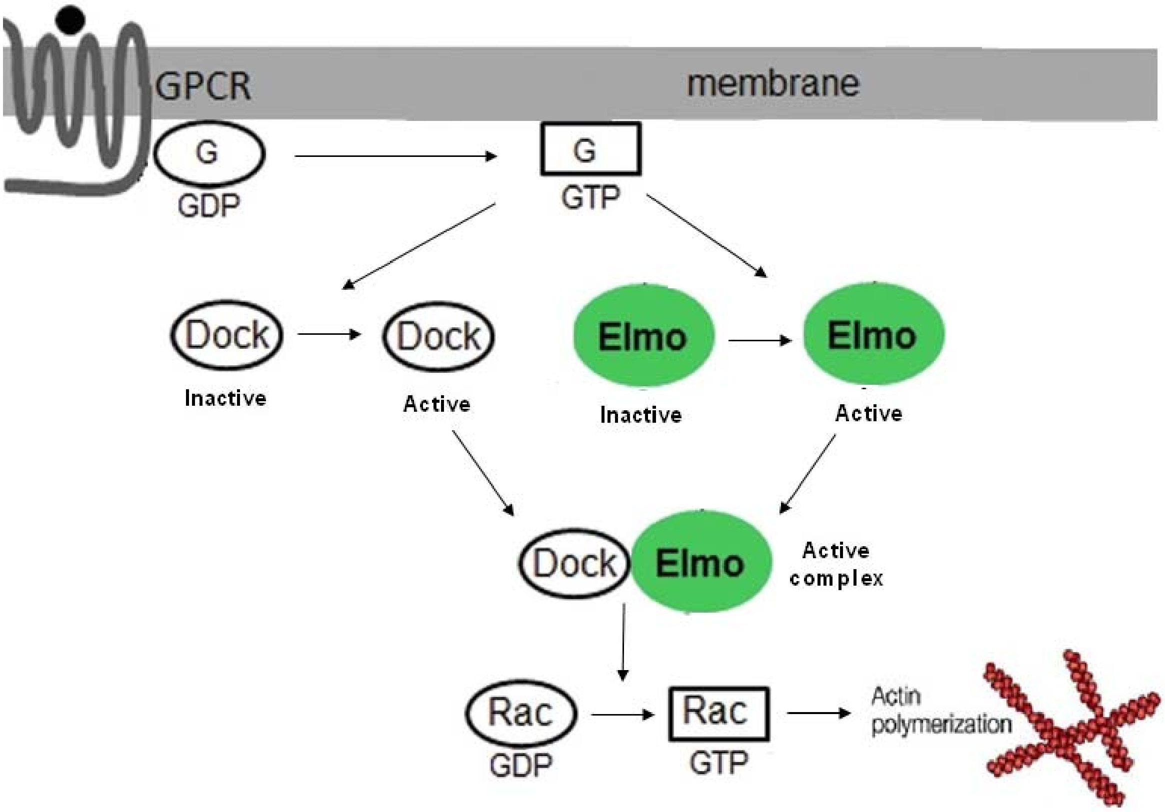


Fig. 2.2

With reference to Fig. 2.2, describe how a chemical signal molecule bound to the GPCR brings about the cellular response.

1. **(Inactive) membrane bound G protein will be activated when its bound GDP is replaced with GTP;**
2. **(active) G protein will activate Elmo and Dock proteins to form a protein complex;**
3. **Elmo-Dock protein complex will activate Rac protein by displacing its GDP with GTP, which will then result in actin polymerisation;**
4. **Ref. to change in 3D configuration of G protein or Rac protein during activation;**

@ 1m, max 3 [3]

[Total: 10]

For examiner's use

3 Fig. 3.1 shows the *arg* operon in *E. coli*, which is a repressible operon with three genes, *arg B, arg C* and *arg H*, that code for enzymes responsible for the biosynthesis of the amino acid arginine. Another gene upstream of the operon (not shown in Fig. 3.1) codes for a repressor protein R, which binds to the operator to regulate the operon. Arginine acts as a corepressor, which activates repressor *R*.

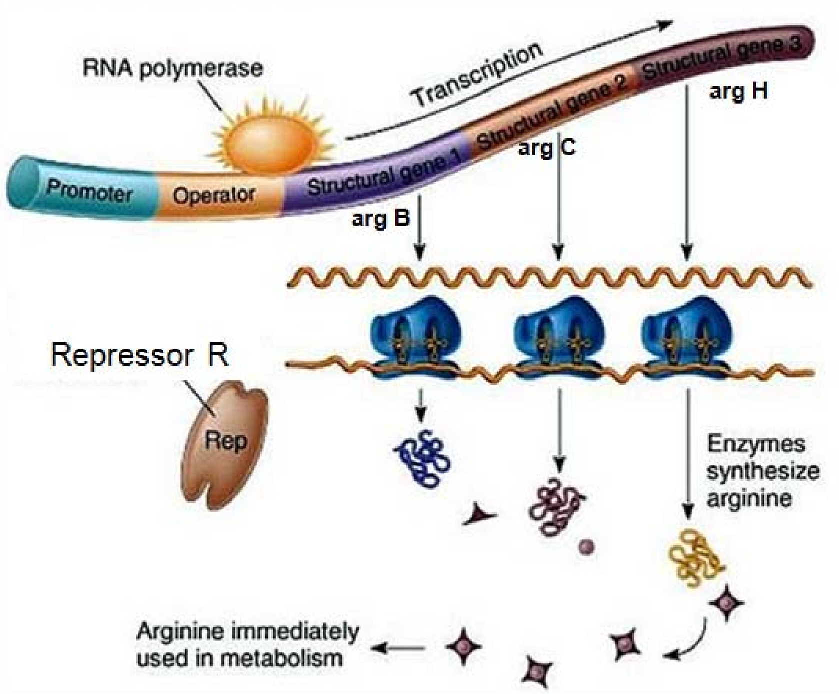


Fig. 3.1

(a) Distinguish between a structural gene and a regulatory gene in the *arg* operon.

1. **A structural gene (is a region of DNA that) codes for a protein / enzyme that synthesises arginine, e.g. *arg* B / *arg* C / *arg* H;**
2. **A regulatory gene codes for a specific protein product, repressor R, that regulates the expression of the structural genes;**

@ 1m [2]

(b) Based on your understanding of a repressible system, suggest how arginine is able to control the level of enzymes involved in arginine synthesis to ensure the conservation of resources by *E. coli*.

1. **When arginine is present, it binds to the allosteric site of repressor R, activating it;**
2. **Repressor R binds to the operator, RNA polymerase is unable to bind to the promoter to transcribe the arg B, arg C and arg H genes;**
3. **Enzymes responsible for the synthesis of arginine are not produced;**
4. **Resources are not wasted to synthesise arginine when it is present in the medium;**

@ 1m [4]

(c) *E. coli* was cultured in a growth medium with all the essential nutrients needed except the amino acid arginine. Fig. 3.2 shows a graph of the concentration of enzymes that are involved in the synthesis of arginine against time. At time **X**, arginine was added to the growth medium. Using an arrow, indicate on Fig. 3.2, the time when arginine was added.

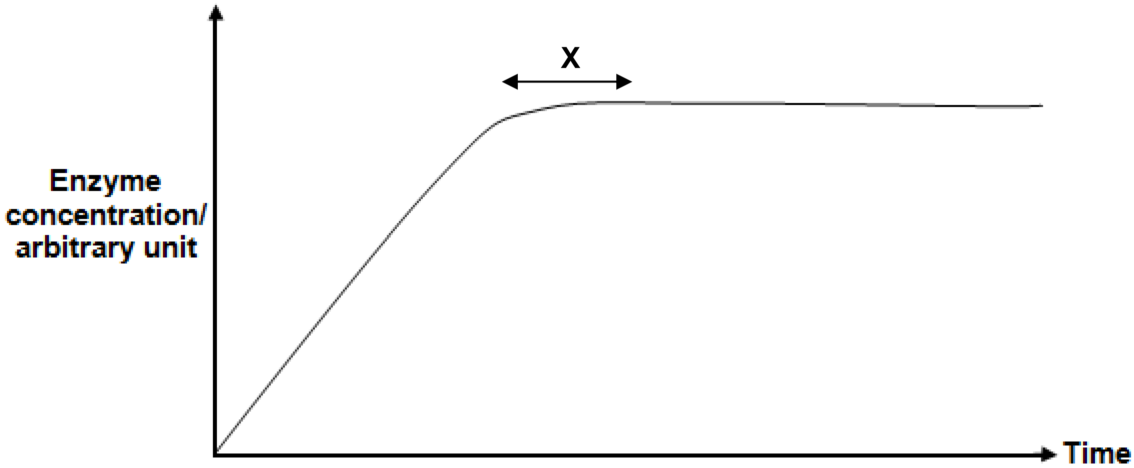


Fig. 3.2

[1]

(d) Besides having operons, prokaryotes have other means to adapt to changing environments through gene transfer.

Describe how viruses can facilitate the transfer of **many** advantageous alleles in a bacterial population.

1. **Phages undergoing lytic life cycle can facilitate generalised transduction;**
2. **During the lytic cycle, bacteria DNA is hydrolysed into fragments;**
3. **Errors may occur in the assembly stage when advantageous alleles of bacteria / DNA fragments are mistakenly packaged into new phage head / OWTTE;**
4. **Phage released can infect another bacterium, introducing the advantageous allele via homologous recombination / integration;**

@ 1m [4]

[Total: 11]

For examiner's use

4 A group of genetics students were examining the inheritance of fruit colour and shape in a particular species of plant. When they crossed a pure breeding plant that had red, oval fruits with pure breeding plants that had yellow, long fruits, all the offspring produced were observed to have red, oval fruits.

They then proceeded to testcross two plants, **A** and **B**, from the F_2 progeny, both with red, oval fruits and the following results were obtained as shown in Table 4.1.

Table 4.1

Phenotype	Progeny of A	Progeny of B
Red, long fruits	46	4
Yellow, oval fruits	44	6
Red, oval fruits	5	43
Yellow, long fruits	5	47

(a) Based on the information and results of the crosses given above, describe the inheritance of fruit colour and shape in this species of plant.

1. **Two genes located on the same chromosome/linked;**
2. **Red allele dominant to yellow allele, and oval dominant to long fruits;**

@ 1m [2]

(b) Using appropriate letters to represent the various alleles, state the genotypes of plant **A** and **B**.

Letter to represent allele for red fruit : **R**

Letter to represent allele for yellow fruit : **r**

Letter to represent allele for oval fruit : **A**

Letter to represent allele for long fruit : **a**

Genotype of plant **A**: (with red, oval fruit) **(Ra)(rA) ;**

Genotype of plant **B**: (with red, oval fruit) **(RA)(ra) ;**

Any appropriate letters and both correct @ 1m

[1]

(c) Draw a genetic diagram to explain the results of the testcross on plant **A** as shown in Table 4.1.

Phenotype of parents: **Red, oval fruits** **x** **yellow, long fruits**

Genotype of parents: **(Ra)(rA)** **x** **(ra)(ra)** **;**

Gametes: (Ra) (rA) (RA) (ra) (ra) (ra) **;**

Recombinants, result of crossing over

F1 genotypes: **(Ra)(ra)** **(rA)(ra)** **(RA)(ra)** **(ra)(ra)** **;**

F1 phenotypes: **Red, long** **Yellow, oval** **Red, oval** **Yellow, long** **;**

Majority, from gametes with no crossing over

Minority, recombinant offspring

- **1 m for correct genotypes of parents in testcross**
- **1 m for correct gametes**
- **1 m for correct F1 genotypes**
- **1 m for correct corresponding phenotypes**
- **1 m for annotations on recombinants (either at the gametes or phenotype level)**

[5]

For examiner's use

(d) Using the results given in Table 4.1, calculate the relative location of the genes for fruit colour and fruit shape.

Crossing over value = [(5 + 5) ÷ (100)] x 100

= 10%;

Correct workings: ½ m
Statement for relative location: ½ m

Distance between the two genes: **10 map units/centimorgans**

[1]

The group of students proceeded to measure the fruit length obtained from true breeding plants with oval fruits and those from true breeding plants with long fruits. Table 4.2 below shows the number of fruits in each fruit length category from the two respective pure breeding plants.

Table 4.2

	Fruit length / cm								
	3	4	5	6		10	11	12	13
Plants with oval fruits	4	7	6	2					
Plants with long fruits						3	6	9	5

(e) Explain the variations observed in fruit length within each group of pure breeding plants.

1. **Variation in fruit length due to environmental factors (such as amount of sunlight (for PS) / location of fruit on plant, number of fruits on same branch / availability of water / nutrients during development);**
2. **Pure breeding plants are homozygous at all gene loci;**

@ 1m [2]

[Total: 11]

For examiner's use

5 *Xenopus* is a genus of aquatic frogs native to sub-Saharan Africa. In a developing *Xenopus* oocyte (egg cell), pre-mRNAs are processed to become mRNAs in the nucleus. They are then exported into the cytoplasm.

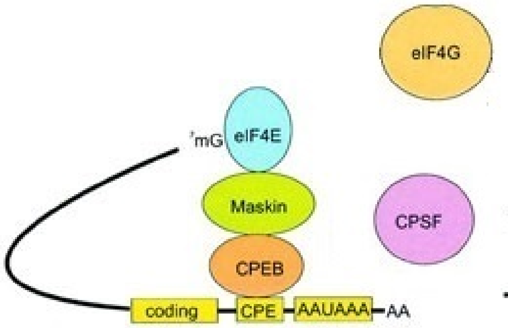
(a) Describe how pre-mRNA is produced in a developing *Xenopus* oocyte.

1. **Ref. to general transcription factors facilitate RNA polymerase binding to / formation of transcription initiation complex at the promoter sequence;**
2. **causing the DNA double helix to unwind and unzip;**
3. **Free ribonucleotides are taken up and matched with the DNA template strand by complementary base pairing;**
4. **Adjacent ribonucleotides are joined together by phosphodiester bonds (by RNA polymerase;**
5. **RNA polymerase transcribes a terminator sequence, releasing the pre-mRNA;**

@ 1m, max 4 [4]

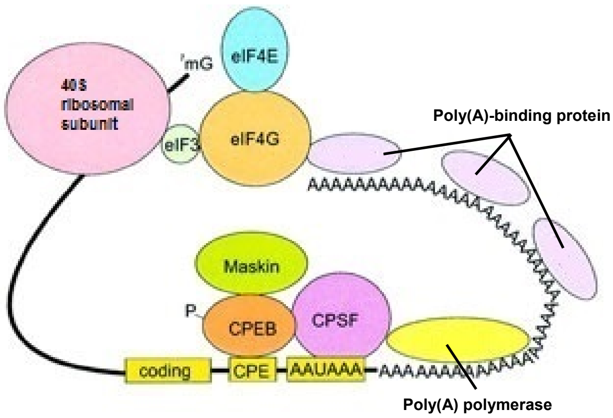
In immature oocytes, these mRNAs are not immediately translated but are stored in the cytoplasm for future use. Fig. 5.1a shows a translationally-repressed mRNA which has a sequence element, CPE, to which CPEB protein binds to. A repressor protein, Maskin binds to the bound CPEB. It then binds to eIF4E, a protein which binds to the modified guanosine cap ('mG) at the 5' end. The binding of Maskin to eIF4E prevents the binding of eIF4G protein to eIF4E, thus inhibiting the formation of the translation initiation complex. CPSF is also a protein involved in the initiation of translation.

Shortly before ovulation, these mRNAs are activated in the oocytes. In Fig. 5.1b, phosphorylation of CPEB protein causes a series of events which leads to the displacement of Maskin from eIF4E, initiating translation.



mRNA in immature oocyte

Fig. 5.1a



mRNA in mature oocyte

Fig. 5.1b

(b) Contrast the structure of mRNAs in an immature oocyte to those in a mature oocyte.

Translationally-repressed mRNAs have shorter poly(A) tails;

@ 1m [1]

(c) With reference to Fig. 5.1a and Fig 5.1b, describe how phosphorylation of CPEB protein activates translation during oocyte maturation.

1. (Phosphorylation of CPEB protein) causes it to bind CPSF, (helping it to associate with the AAUAAA sequence);
2. CPSF recruits poly(A) polymerase to the 3' end of mRNA where it will add more adenine nucleotides to the poly(A) tail / catalyses poly(A) addition;
3. The newly elongated poly(A) tail is then bound by poly(A)-binding protein which in turn associates with eIF4G;
4. eIF4G then displaces Maskin and binds to eIF4E;
5. (Through eIF3, eIF4G interacts with) the small 40S ribosomal subunit binds with the 5' end of mRNA to initiate translation;

@1m, max 4 [4]

(d) Describe how degradation of Maskin occurs in the cell when it is no longer needed by the cell.

1. Enzymes in the cytoplasm attach ubiquitin molecules to Maskin;
2. Proteasome (recognises and) degrades the ubiquinated Maskin into short peptides;
3. ATP is expended in the process;

@ 1m [3]

(e) Suggest the advantage of control of gene expression at the level of translation over control of gene expression at the level of transcription.

1. Ref. to faster formation of protein in the cell (as not needed to activate transcription of gene)
2. Possible even if nucleus is absent from cell;
3. Provides a mean for cell to cope with rapid changes in protein concentrations;

@ 1m [1]

[Total: 13]

For examiner's use

6 (a) Trinucleotide repeat expansion is a type of mutation that inserts and increases the number of base triplets in the DNA sequence. Although trinucleotide repeat expansion does not result in frameshift mutation, it can have detrimental effects on the structure and function of protein.

Fig. 6.1 illustrates an example of trinucleotide repeat expansion in the coding sequence of a gene and its effect on the polypeptide produced.

-- CAT TCA CAG GTA ATC ATG CTA --
His Ser Gln Val Ile Met Leu

↓

-- CAT TCA CAG GTA GTA GTA GTA ATC ATG CTA --
His Ser Gln Val Val Val Val Ile Met Leu

Fig. 6.1

(i) Describe the difference between trinucleotide repeat expansion and gene amplification in their effects on gene expression.

1. Normal amounts of polypeptide vs. increased numbers of polypeptide;

2. Normal rate of transcription/ translation/protein synthesis vs increased rate of transcription/translation/protein synthesis; A! RNA/mRNA

3. No change on the primary structure vs changes the primary structure;

@ 1m [1]

(ii) Explain how trinucleotide repeat expansion could adversely affect the enzyme protease.

1. **Increased numbers of base triplets results in extra numbers of a.a. in the polypeptide sequence;**

2. **Ref. to the types of bonds disrupted/newly formed between R groups of a.a. e.g. H, ionic bonds, hydrophobic interactions; (A! one example)**

3. **3D configuration of protease is disrupted;**

4. **Ref. to the (protein) substrate is no longer complementary to the active site to form an Enzyme-substrate complex;**

5. ***Protease can no longer break down / hydrolyse protein into peptides / break peptide bonds**

@ 1m, * compulsory point

[4]

(b) The huntingtin gene codes for the huntingtin (Htt) protein found in the cytoplasm. Huntington's disease is a neurodegenerative disorder caused by the increase in CAG (a base triplet coding for the amino acid glutamine) repeat in the mutated huntingtin gene.

The resultant mutant huntingtin (mHtt) protein contains an abnormally long stretch of glutamine residues, which results in the mHtt protein aggregating and accumulating within the neurones as shown in Fig. 6.2a. Fig. 6.2b shows the molecular structure of the amino acid glutamine.

Fig. 6.2a

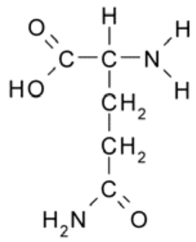


Fig. 6.2b

(i) Many globular proteins are soluble in water as the hydrophobic amino acids are buried in the interior while the hydrophilic amino acids are found on the exterior.

Based on Fig. 6.2a, Fig. 6.2b and the information above, suggest and explain why the mutant huntingtin protein molecules containing abnormally long stretch of glutamine residues aggregate and become insoluble in water.

1. Ref. to the polar/hydrophilic <u>R group</u> of glutamine;
2. Large numbers of glutamine residues increases the cross-links by hydrogen bonds formed <u>between</u> the mHtt protein molecules;
3. No hydrogen bonds formed between water molecules and glutamine on mHtt protein;

@ 1m [3]

(ii) The mutant huntingtin protein has been discovered to bind with and disrupt the function of complex II (an electron carrier) embedded in the inner membrane of mitochondria. Using your knowledge of respiration, explain how the mutant huntingtin protein will change the pH in the intermembrane space of mitochondria.

1. *pH will increase;
2. Since mHtt protein inhibits electron transfer by the electron carriers (down the energy levels) along ETC;
3. No energy is harnessed by electron carriers / proton pumps (A! complexes I, III and IV) to pump protons from mitochondrial matrix into intermembrane space;
4. Proton gradient/pool is no longer formed;

@ 1m, *compulsory [3]

[Total: 11]

For examiner's use

7 Primate species have the potential to visually perceive three colour spectrums – blue, green and red.

Fig. 7.1 shows a phylogenetic tree of primate evolution. The common ancestor for all primate species was thought to be dichromatic, being able to perceive blue and one other colour only (hence unable to distinguish green and red colour). Only the species within the group of "apes" and "old world monkeys" are found to be consistently trichromatic and are able to perceive all three colours.

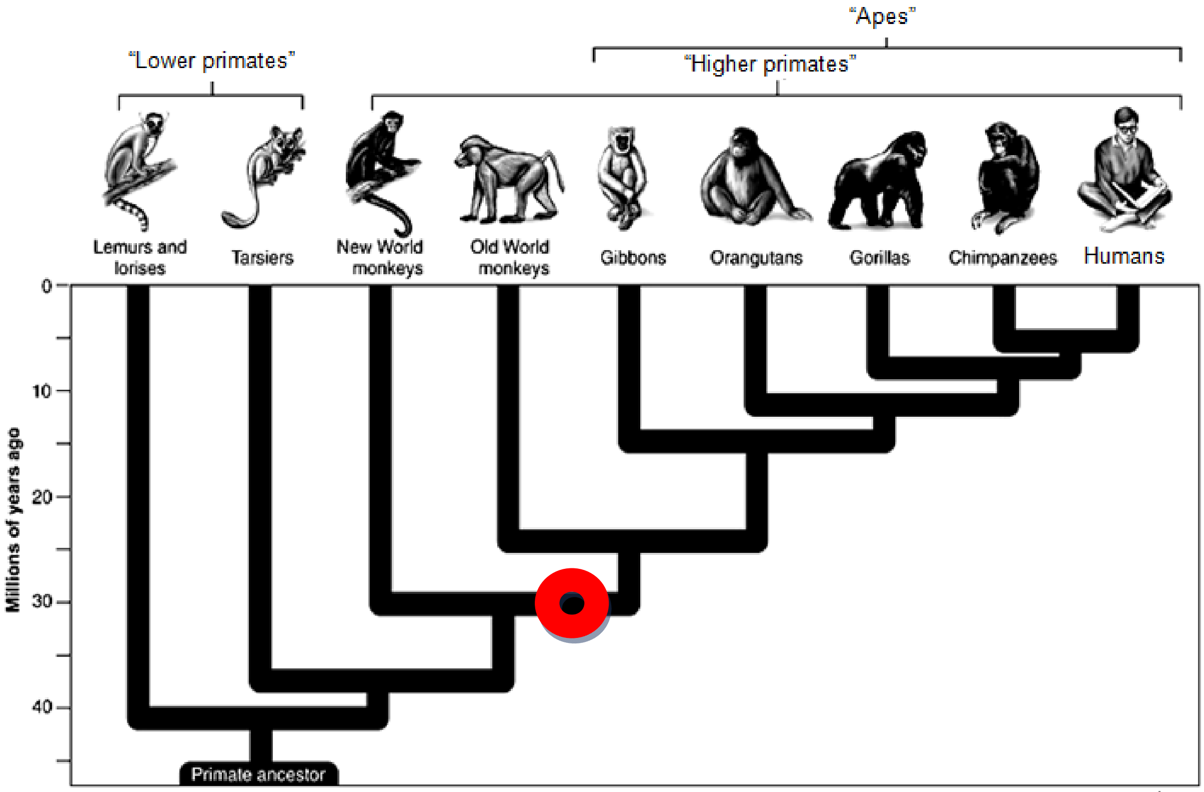


Fig. 7.1

(a) **(i)** Using an arrow, indicate on Fig. 7.1 the point where trichromatic vision is likely to have evolved in primates. [1]

(ii) Suggest how trichromatic vision may be of survival value to fruit-feeding primates in forest habitats.

Allow primates to distinguish (ripened) fruits from green background;

@ 1m [1]

For examiner's use

(iii) Explain how natural selection has resulted in the evolution of trichromatic vision in a population.

1.	**Individuals with this trait would have a selective advantage / be selected for (because they are better able to obtain food);**
2.	**More survive to reproduce / experience higher reproductive success / have greater fitness;**
3.	**Genes coding for the trait will be passed down to subsequent generations;**
4.	***Over time, this leads to an increase in allele frequency for the trait in the population;**

@ 1m, pt 4 compulsory, max 3 [3]

Howler monkeys (*Alouatta* spp.) and night monkeys (*Aotus* spp.) are two genera belonging to the group of "new world monkeys". Both genera of monkeys are native to South and Central American forests. Howler monkeys are diurnal (active in the day) and are known for communicating by loud "howls". Night monkeys, however, are nocturnal (active at night) and communicate by owl-like "hoots".

(b) Explain how the two genera of monkeys could have evolved from a common ancestor.

1. **Ref. to sympatric speciation / new species evolved within the same geographical area as ancestral species;**
2. **Ref. to temporal isolation / behavioural isolation / physiological isolation;**
3. **Due to reduced gene flow, subpopulations evolve (divergently) to form different species;**
4. **Individuals of the two subpopulations no longer able to interbreed to produce fertile, viable offspring / Ref. to reproductive isolation established;**

@ 1m [4]

All known species of "new world monkeys" have dichromatic vision, with the exception of howler monkeys which have evolved trichromatic vision.

(c) Discuss whether trichromatic vision shared by both howler monkeys and humans is an example of a homologous trait.

1. **No / It is not an example of homology / It is an example of analogy instead;**
2. **Trichromatic vision in Howler monkeys evolved independently / is not inherited from the same <u>recent</u> common ancestor as Humans;**
3. **It evolved due to similar selection pressures (of the need to identify food) in the environment / due to convergent evolution;**

@ 1m [3]

[Total: 12]

For examiner's use

Section B

Answer **EITHER 8 OR 9.**

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

EITHER

8 **(a)** Describe how the structure of a lysosome is adapted for its function. [8]

Structural Features	Function
1. Membrane bound organelle / ref to phospholipid bilayer / impermeable to protons	**2. To maintain different chemical environment (acidic environment);** **3. To prevent hydrolytic enzyme from interfering with other processes in the cytoplasm;**
4. Allow high concentration of enzymes to accumulate;	**5. for high rate of hydrolysis;**
6. Presence of proton pump embedded on phospholipid bilayer of lysosome;	**7. Active transport of H^+ into lysosome;** **8. Create acidic environment for hydrolytic enzymes / optimal pH of hydrolytic enzyme is acidic;**
9. Presence of hydrolytic enzymes in lysosome;	**10. For digestion of ingested particles;** **11. For break down of defunct or surplus organelle / autophagy;** **12. For <u>autolysis</u> of cell;**
13. Fluidity of membrane;	**14. Allow lysosomal membrane to fuse with endocytic /phagocytic vesicle;** **15. Rupture of membrane to release hydrolytic enzymes;**
16. modification of molecules in membrane;	**17. prevent digestion of membrane;**
18. presence of carrier/ transporter on lysosome membrane;	**19. transport products of digestions back into cytosol;**

(b) Explain the production of a small yield of ATP from anaerobic respiration in mammalian cells. [6]

1. Ref. to incomplete/partial oxidation of glucose during glycolysis in the absence of oxygen;
2. 1 molecule of glucose is broken down to 2 molecules of pyruvate;
3. NAD^+ is required to oxidise glucose;
4. With the net yield of 2 ATP and 2 reduced NAD;
5. Ref. to ATP production by <u>substrate level phosphorylation</u>;
6. Pyruvate will be <u>reduced</u> to lactate by accepting H atoms from reduced NAD;
7. Ref. to lactate dehydrogenase;
8. NAD^+ is regenerated so that glycolysis can continue to occur (to yield small amount of ATP);

@ 1m

For examiner's use

(c) Contrast the products of meiosis with those of mitosis and explain how these differences come about. [6]

Products of mitosis	Products of meiosis	Explanation
1. Genetically identical daughter cells vs	**Non-identical daughter cells formed;**	**2. Due to the process of crossing over between non-sister chromatids of homologous chromosomes / at bivalent during Prophase I;** **3. Independent assortment of homologous chromosomes at metaphase I / chromatids at metaphase II;**
4. Number of chromosomes identical to parent cell vs	**Number of chromosomes halved of the original number in parent cell;**	**5. Due to segregation of homologous chromosomes during anaphase I;** **6. Semi-conservative replication of DNA occurs only once during Interphase I;**
7. Two daughter cells formed vs	**Four daughter cells formed;**	**8. Two nuclear / cytoplasmic divisions during the process of meiosis vs one nuclear / cytoplasmic division in mitosis;**

[20]

For examiner's use

9 **(a)** Describe the types of bonding which hold a protein in shape. [6]

1. **The bonds may be formed between different polypeptides or between different parts of the same polypeptide;**
2. **Ionic bond is formed between acidic and basic R groups of amino acid/ the electrostatic attraction between two oppositely-charged R groups;**
3. **Disulphide bond is formed between two molecules of cysteine line up alongside each other/ OWTTE;**
4. **Hydrophobic interactions between amino acids with non-polar R-groups;**
5. **Hydrogen bond is formed between amino acids with polar R-groups;**
6. **Ref to folding into 3D configuration/ globular shape/ fibrous shape;**
7. **Hydrogen bond is formed between –CO and –NH groups (of two peptide bonds) of the polypeptide backbone;**
8. **Ref to producing α-helix and β-pleated sheet;**

@ 1m

(b) Explain how genotype is linked to phenotype. [8]

<u>Definition of genotype and phenotype</u>

1. **A genotype of an organism is determined by the alleles of specific genes; A! genetic make-up with respect to the alleles**
2. **The phenotype of an organism is the physical manifestation / trait resulting from a specific genotype;**
3. **And its interaction with the environment;**

<u>Explanation of how genotype is linked to phenotype</u>

4. **Ref. to when different alleles are transcribed to form mRNA;**
5. **Ref. to mRNA is translated into polypeptide chain(s) / protein(s);**
6. **Different alleles will result in the production of different polypeptide chain(s) / protein(s);**
7. **Different gene products / proteins may affect different cellular pathways / different parts of a pathway / different functions;**
8. **Resulting in different phenotypes;**

<u>Effect of dominant and recessive alleles</u>

9. **The alleles present for each particular gene may be dominant or recessive;**
10. **Dominant alleles masks the effect / influence of recessive alleles;**
11. **Hence only the dominant allele will be expressed in the phenotypes of homozygous dominant and heterozygous organisms;**
12. **Phenotypes of homozygous recessive individuals will only be expressed when both alleles are recessive;**

@ 1m

For examiner's use

(c) With reference to the stages of cell signalling, describe the response of liver cells to insulin. [6]

	Stages	Insulin
1.	**Reception**	**The binding of insulin (to receptor tyrosine kinase) causes two receptor monomers to aggregate and form an dimer;**
2.		**This activates the tyrosine kinase region of each polypeptide, which uses ATP to cross-phosphorylate the tyrosine residues on each other's cytoplasmic tails.**
3.	**Transduction**	**Activation of relay proteins;**
4.		**Generation of 2nd messengers e.g. calcium ions;**
5.	**Response**	**Increases / activates enzymes involved in formation of glycogen from glucose / glycogenesis;**
6.		**Inhibits glycogenolysis and gluconeogenesis;**
7.		**Increases uptake of amino acids for the synthesis of amino acids / uptake of fatty acids for the synthesis lipids;**
8.		**Increases rate of respiration;**
9.		**Increases rate of DNA and RNA synthesis;**

@ 1m, max 3 for response

[20]